IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

<Name>
<Date>

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Summary of methodologies**

As part of this study, several methodological steps were followed to ensure the quality and reliability of the results obtained:

1. Data collection and preparation;
2. Exploratory data analysis;
3. Model building and training;
4. Validation and evaluation.

- Summary of all results

- **Summary of all results**

The prepared data showed sufficient quality for model training.

Exploratory analyses highlighted the most influential variables as well as the main trends in the dataset.

The different models tested showed variable performance depending on their parameters and complexity.

Hyperparameter optimization using techniques such as cross-validation and grid search allowed overall performance to be improved.

The selected model is the one that achieved the best metrics on the validation and test sets, demonstrating good generalization capability.

The results obtained confirm the relevance of the adopted methodological approach and provide a solid basis for future use or work.

Introduction

- Project background and context

The rapid growth of digital data has made Machine Learning a key tool for extracting insights and supporting decision-making. This project applies supervised learning techniques, including Logistic Regression and Support Vector Machines (SVM), through data preprocessing, model training, optimization, and evaluation. The objective is to identify the most effective classification model and support data-driven decision-making.

- Problems you want to find answers

1. How can the available dataset be effectively prepared and processed for machine learning applications?
2. Which features have the greatest influence on the target variable?
3. How well do different classification algorithms perform on the dataset?
4. Can model performance be improved through hyperparameter tuning and cross-validation techniques?
5. Which model provides the best balance between accuracy, robustness, and generalization on unseen data?
6. What insights can be derived from the results to support future decision-making and predictive analysis?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - Describe how data was collected
- Perform data wrangling
  - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - How to build, tune, evaluate classification models

Data Collection

- Describe how data sets were collected.

The dataset was obtained from a publicly available data source and collected for machine learning analysis. The collected data were inspected to identify missing values, inconsistencies, and duplicates. After cleaning and preprocessing, the dataset was validated to ensure quality and reliability. The final dataset was then prepared and divided into training and testing sets for model development and evaluation.

- You need to present your data collection process use key phrases and flowcharts

Data source => Data collection => Data Inspection => Data Cleaning & Preprocessing => Data Validation => Dataset ready => Train/Test Split => Model Training & Evaluation

Data Collection – SpaceX API

- Present your data collection with SpaceX REST calls using key phrases and flowcharts

- Add the GitHub URL of the completed SpaceX API calls notebook (must include completed code cell and outcome cell), as an external reference and peer-review purpose

SpaceX REST API => API REST Calls => JSON Data Retrieval => Data Extraction => Data Transformation => Data Cleaning => Data Validation => Final Dataset => Analysis & Modeling

URL

https://github.com/dafe-hi/Data-Science-Notebook.git

Data Collection - Scraping

- Present your web scraping process using key phrases and flowcharts

- Add the GitHub URL of the completed web scraping notebook, as an external reference and peer-review purpose

Target Website => HTTP Request => HTML Page Retrieval => HTML Parsing => Data Extraction => Data Cleaning => Data Transformation => Data Validation => Final Dataset => Analysis & Modeling

URL

https://github.com/dafe-hi/Data-Science-Notebook.git

Data Wrangling

- Describe how data were processed

Data wrangling transformed raw data from the SpaceX REST API and web scraping sources into a clean, structured dataset. The process included data integration, cleaning, handling missing values, removing duplicates, and feature engineering. The final validated dataset was prepared for exploratory analysis and machine learning modeling.

- You need to present your data wrangling process using key phrases and flowcharts

Raw Data Sources (API + Web Scraping) => Data Integration => Missing Values Handling => Duplicate Removal => Data Cleaning => Data Transformation => Feature Engineering => Data Validation => Final Dataset => Analysis & Modeling

- Add the GitHub URL of your completed data wrangling related notebooks, as an external reference and peer-review purpose

https://github.com/dafe-hi/Data-Science-Notebook.git

EDA with Data Visualization

- Summarize what charts were plotted and why you used those charts

1. Relationship between Flight Number and Launch Site

A scatter plot helps visualize how launches are distributed across different launch sites and flight numbers.

2. Relationship between Payload Mass and Launch Site

A scatter plot allows comparison of payload masses launched from different sites and their corresponding outcomes.

3. Relationship between Success Rate and Orbit Type

A bar chart provides a clear comparison of success rates across orbit types.

4. Relationship between Flight Number and Orbit Type

A scatter plot helps identify trends in orbit selection throughout the launch history.

5. Relationship between Payload Mass and Orbit Type

A scatter plot is useful for examining how payload mass varies among orbit types.

6. Launch Success Yearly Trend

A line chart effectively displays changes in launch success rates over time.

- Add the GitHub URL of your completed EDA with data visualization notebook, as an external reference and peer-review purpose

https://github.com/dafe-hi/Data-Science-Notebook.git

EDA with SQL

- Using bullet point format, summarize the SQL queries you performed

%sql SELECT DISTINCT Launch_Site FROM SPACEXTBL;

%sql SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5;

%sql SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass FROM SPACEXTBL WHERE CUSTOMER = 'NASA (CRS)';

%sql SELECT AVG(PAYLOAD_MASS__KG_) AS Average_Payload_Mass FROM SPACEXTBL WHERE BOOSTER_VERSION = 'F9 v1.1';

%sql SELECT MIN(Date) AS First_Successful_Ground_Pad_Landing FROM SPACEXTBL WHERE Landing_Outcome = 'Success (ground pad)';

%sql SELECT DISTINCT Booster_Version FROM SPACEXTBL WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;

%sql SELECT Mission_Outcome, COUNT(*) AS Total_Missions FROM SPACEXTBL GROUP BY Mission_Outcome;

%sql SELECT Booster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL);

%sql SELECT substr(Date, 6, 2) AS Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTBL WHERE substr(Date, 1, 4) = '2015' AND Landing_Outcome LIKE '%Failure (drone ship)%';

%sql SELECT Landing_Outcome, COUNT(*) AS Outcome_Count FROM SPACEXTBL WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Outcome_Count DESC;

- Add the GitHub URL of your completed EDA with SQL notebook, as an external reference and peer-review purpose

https://github.com/dafe-hi/Data-Science-Notebook.git

Build an Interactive Map with Folium

- Summarize what map objects such as markers, circles, lines, etc. you created and added to a folium map
- Explain why you added those objects
- Add the GitHub URL of your completed interactive map with Folium map, as an external reference and peer-review purpose

Build a Dashboard with Plotly Dash

- Summarize what plots/graphs and interactions you have added to a dashboard
- Explain why you added those plots and interactions

1. Pie Chart – Launch Success Distribution

This chart provides a quick overview of successful launches by site and helps compare launch site performance.

2. Scatter Plot – Payload Mass vs. Launch Outcome

This visualization helps determine whether payload mass influences mission success or failure.

3. Scatter Plot – Payload Mass vs. Launch Site

This chart allows comparison of payload distributions across different launch sites.

4. Scatter Plot – Flight Number vs. Launch Site

This visualization highlights launch frequency and performance trends over time at each site.

5. Bar Chart – Success Rate by Orbit Type

This chart clearly compares the success rates of different orbit categories and identifies the most successful mission types.

6. Scatter Plot – Flight Number vs. Orbit Type

This graph illustrates how orbit types evolved throughout the launch history and mission development.

7. Scatter Plot – Payload Mass vs. Orbit Type

This visualization helps identify relationships between payload weight and destination orbit.

8. Line Chart – Yearly Launch Success Trend

This chart shows how SpaceX's launch success rate changed over time and highlights improvements in operational reliability.

- Add the GitHub URL of your completed Plotly Dash lab, as an external reference and peer-review purpose

https://github.com/dafe-hi/Data-Science-Notebook.git

Predictive Analysis (Classification)

- Summarize how you built, evaluated, improved, and found the best performing classification model

The model development process involved data preparation, feature selection, and splitting the dataset into training and testing sets. Multiple classification algorithms (Logistic Regression, SVM, Decision Tree, and KNN) were trained and optimized using Grid Search and cross-validation. Their performances were compared using accuracy metrics, and the best-performing model was selected to predict Falcon 9 first-stage landing success.

- You need present your model development process using key phrases and flowchart

Prepared Dataset => Feature Selection => Train-Test Split => Build Models (LR, SVM, DT, KNN) => Model Evaluation => Hyperparameter Tuning (GridSearch) => Cross Validation => Compare Accuracy => Select Best Model => Landing Prediction

- Add the GitHub URL of your completed predictive analysis lab, as an external reference and peer-review purpose

https://github.com/dafe-hi/Data-Science-Notebook.git

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Section 2

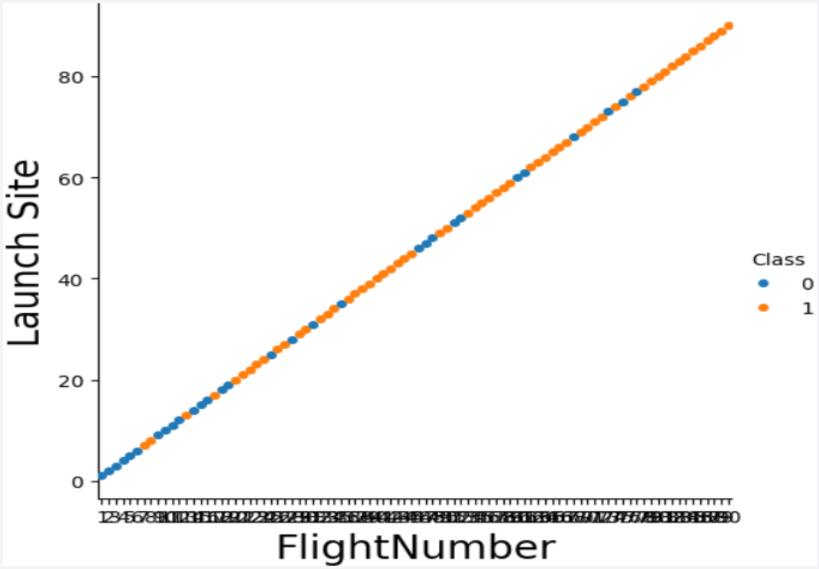
Insights drawn from EDA

Flight Number vs. Launch Site

- Show a scatter plot of Flight Number vs. Launch Site

```
sns.catplot(y="FlightNumber", x="FlightNumber", hue="Class", data=df)
plt.xlabel("FlightNumber",fontsize=20)
plt.ylabel("Launch Site",fontsize=20)
plt.show()
```

- Show the screenshot of the scatter plot with explanations

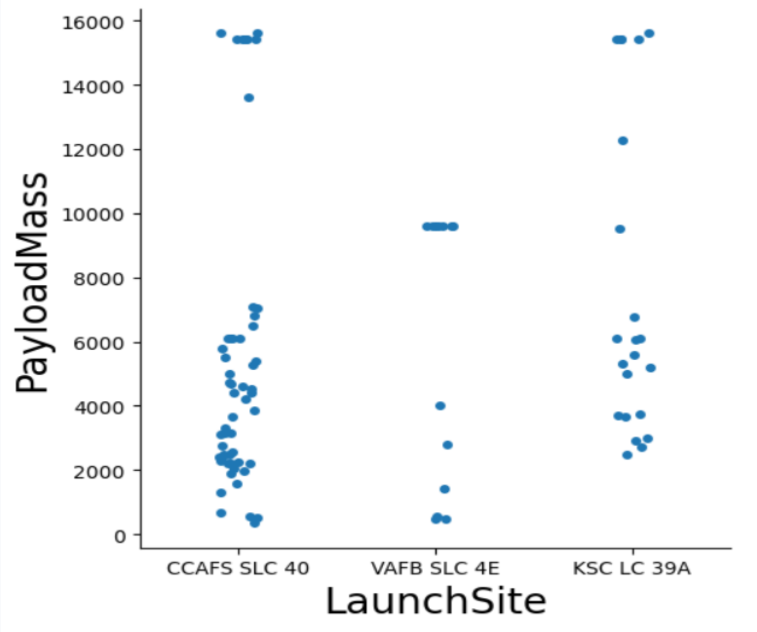


Payload vs. Launch Site

- Show a scatter plot of Payload vs. Launch Site

```
sns.catplot(y="PayloadMass", x="LaunchSite", hue="Class", data=df)
plt.xlabel("LaunchSite",fontsize=20)
plt.ylabel("PayloadMass",fontsize=20)
plt.show()
```

- Show the screenshot of the scatter plot with explanations

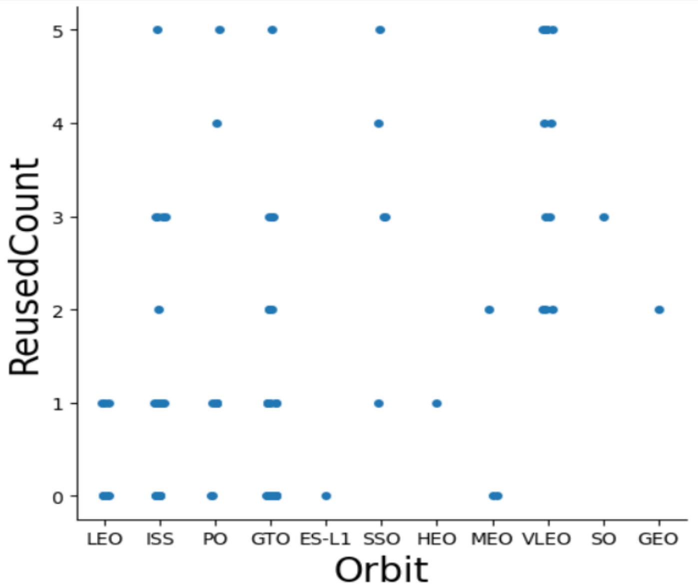


Success Rate vs. Orbit Type

- Show a bar chart for the success rate of each orbit type

```
sns.catplot(y="ReusedCount", x="Orbit", data=df)
plt.xlabel("Orbit",fontsize=20)
plt.ylabel("ReusedCount",fontsize=20)
plt.show()
```

- Show the screenshot of the scatter plot with explanations



Flight Number vs. Orbit Type

- Show a scatter point of Flight number vs. Orbit type

Plot a scatter point chart with x axis to be FlightNumber and y axis to be the Orbit, and hue to be the class value

HINT use groupby method on Orbit column and get the mean of Class

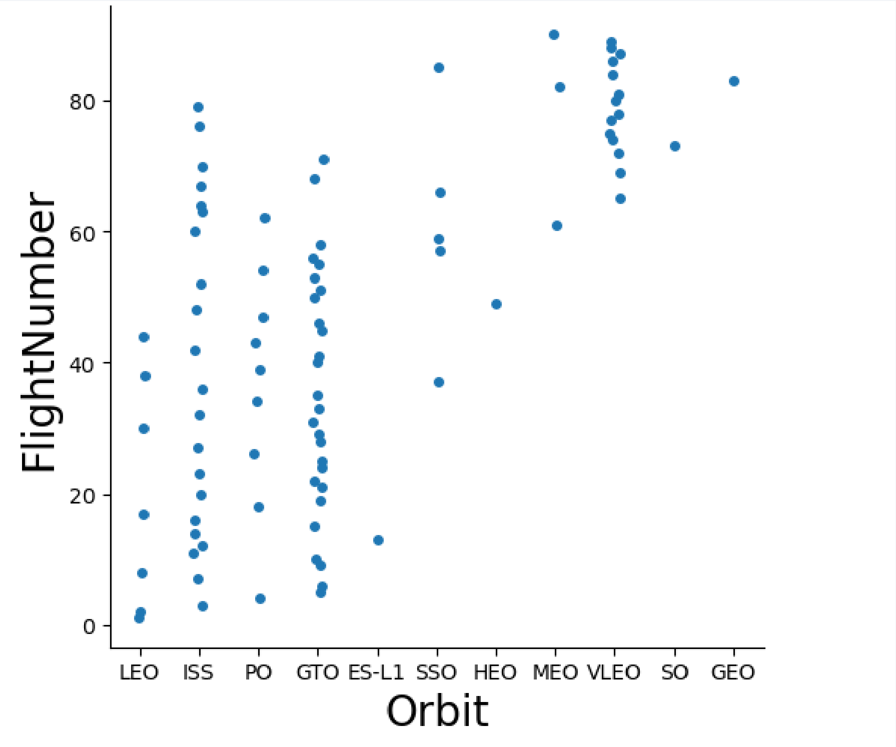
columnsns.catplot(y="FlightNumber", x="Orbit", data=df)

plt.xlabel("Orbit",fontsize=20)

plt.ylabel("FlightNumber",fontsize=20)

plt.show()

- Show the screenshot of the scatter plot with explanations

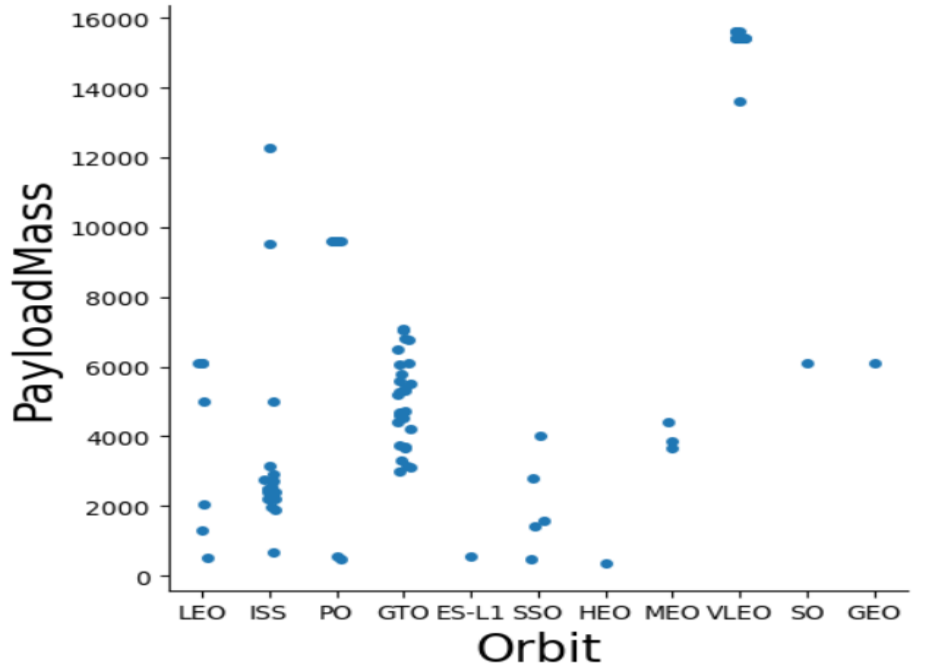


Payload vs. Orbit Type

- Show a scatter point of payload vs. orbit type

```
sns.catplot(y="PayloadMass", x="Orbit", data=df)
plt.xlabel("Orbit",fontsize=20)
plt.ylabel("PayloadMass",fontsize=20)
plt.show()
```

- Show the screenshot of the scatter plot with explanations

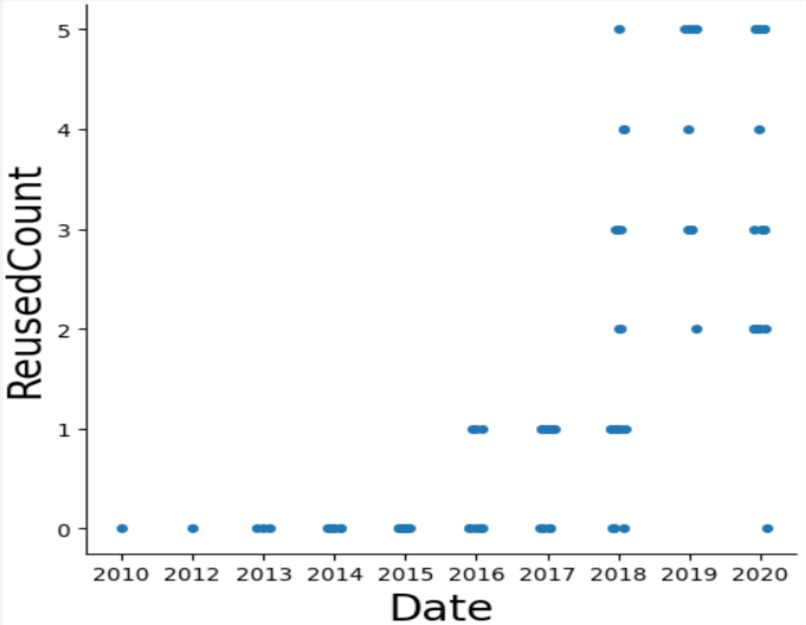


Launch Success Yearly Trend

- Show a line chart of yearly average success rate

```
sns.catplot(y="ReusedCount", x="Date", data=df)
plt.xlabel("Date",fontsize=20)
plt.ylabel("ReusedCount",fontsize=20)
plt.show()
```

- Show the screenshot of the scatter plot with explanations



All Launch Site Names

- Find the names of the unique launch sites

```
%sql SELECT DISTINCT Launch_Site FROM SPACEXTBL;
```

- Present your query result with a short explanation here

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with `CCA`

%sql SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5;

- Present your query result with a short explanation here

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Calculate the total payload carried by boosters from NASA

%sql SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass FROM SPACEXTBL WHERE CUSTOMER = 'NASA (CRS)';

- Present your query result with a short explanation here

Total_Payload_Mass
45596

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1

```
%sql SELECT AVG(PAYLOAD_MASS__KG_) AS Average_Payload_Mass FROM SPACEXTBL WHERE BOOSTER_VERSION = 'F9 v1.1';
```

- Present your query result with a short explanation here

verage_Payload_Mass
2928.4

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad

```
%sql SELECT MIN(Date) AS First_Successful_Ground_Pad_Landing FROM SPACEXTBL WHERE Landing_Outcome = 'Success (ground pad)';
```

- Present your query result with a short explanation here

First_Successful_Ground_Pad_Landing
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

```
%sql SELECT DISTINCT Booster_Version FROM SPACEXTBL WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;
```

- Present your query result with a short explanation here

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes

```
%sql SELECT Mission_Outcome, COUNT(*) AS Total_Missions FROM SPACEXTBL GROUP BY Mission_Outcome;
```

- Present your query result with a short explanation here

Mission_Outcome	Total_Missions
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass

```
%sql SELECT Booster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL);
```

- Present your query result with a short explanation here

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

```
%sql SELECT substr(Date, 6, 2) AS Month, Landing_Outcome, Booster_Version, Launch_Site FROM
SPACEXTBL WHERE substr(Date, 1, 4) = '2015' AND Landing_Outcome LIKE '%Failure (drone ship)%';
```

- Present your query result with a short explanation here

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

```
%sql SELECT Landing_Outcome, COUNT(*) AS Outcome_Count FROM SPACEXTBL WHERE Date BETWEEN
'2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Outcome_Count DESC;
```

- Present your query result with a short explanation here

Landing_Outcome	**Outcome_Count**
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

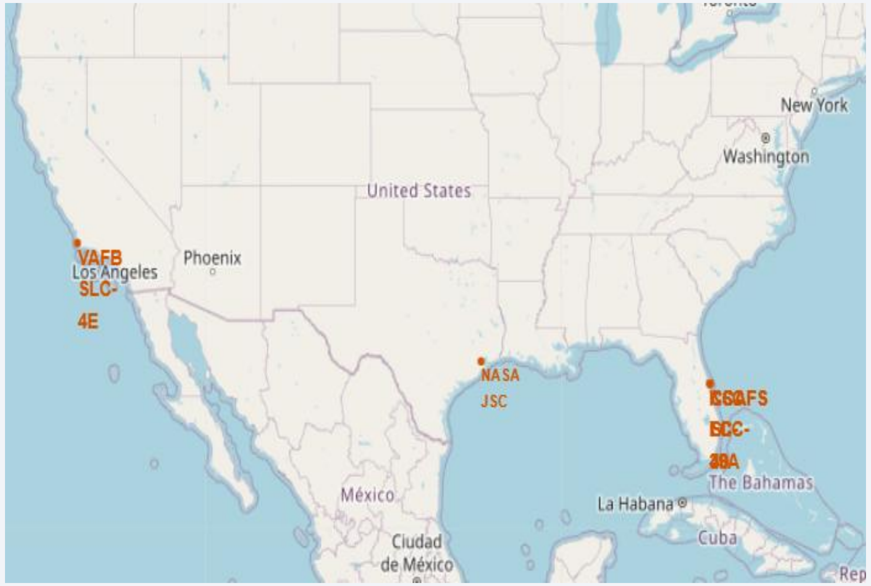
Launch Sites Proximities Analysis

<Folium Map Screenshot 1>

- Replace <Folium map screenshot 1> title with an appropriate title

Geographical Location of all launch sites

- Explore the generated folium map and make a proper screenshot to include all launch sites' location markers on a global map



- **Explain the important elements and findings on the screenshot**

  Important Elements
  Interactive Map
  Highlighted Circle
  Marker Label
  Popup Information
  Geographic Coordinates

<Folium Map Screenshot 2>

- Replace <Folium map screenshot 2> title with an appropriate title

Geographical representation of launch results

- Explore the folium map and make a proper screenshot to show the color-labeled launch outcomes on the map

- Explain the important elements and findings on the screenshot

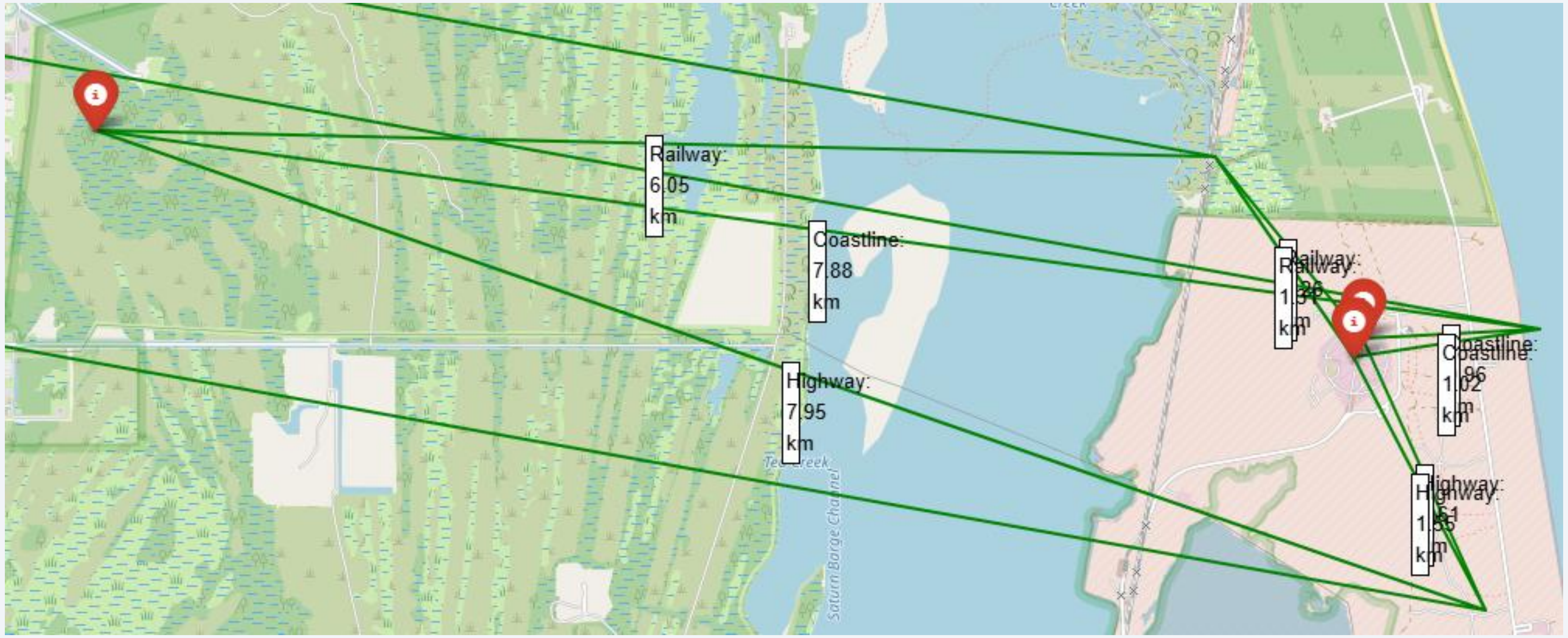
The map shows launch sites with color-coded outcomes: green for successful launches and red for failed launches. The distribution of these markers helps compare site performance, with more green markers indicating higher success rates and more red markers indicating more failures.

<Folium Map Screenshot 3>

- Replace <Folium map screenshot 3> title with an appropriate title

Enable coordinate tracking to identify proximity locations around launch sites

- Explore the generated folium map and show the screenshot of a selected launch site to its proximities such as railway, highway, coastline, with distance calculated and displayed



- Explain the important elements and findings on the screenshot

Launch Site Marker
Interactive Map;
MousePosition Tool;
Nearby Infrastructure

The MousePosition tool allows precise identification of nearby infrastructure coordinates. These coordinates are used to calculate distances from the launch site to railways, highways, and coastlines, helping assess environmental and safety factors. Proximity to the coastline is especially important, as launch sites are often located near the ocean to minimize risks to populated areas.

Section 4

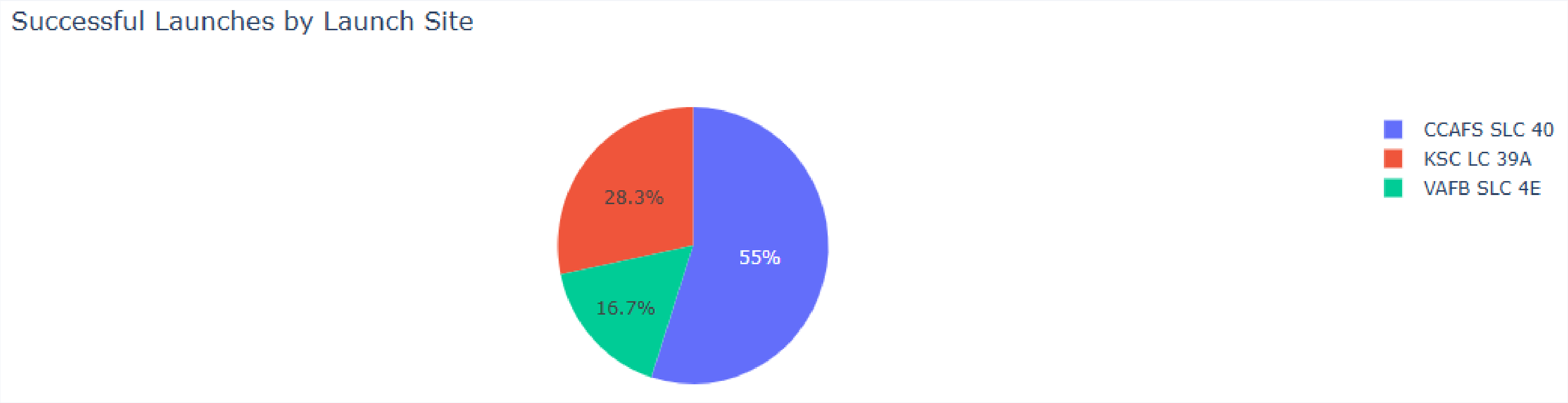
Build a Dashboard with Plotly Dash

<Dashboard Screenshot 1>

- Replace <Dashboard screenshot 1> title with an appropriate title

Launch Success Count Distribution Across All Launch Sites

- Show the screenshot of launch success count for all sites, in a piechart



- Explain the important elements and findings on the screenshot

  Pie Chart Visualization

  Launch Site Categories

  Slice Size

  Labels and Percentages

  Chart Title

  Findings

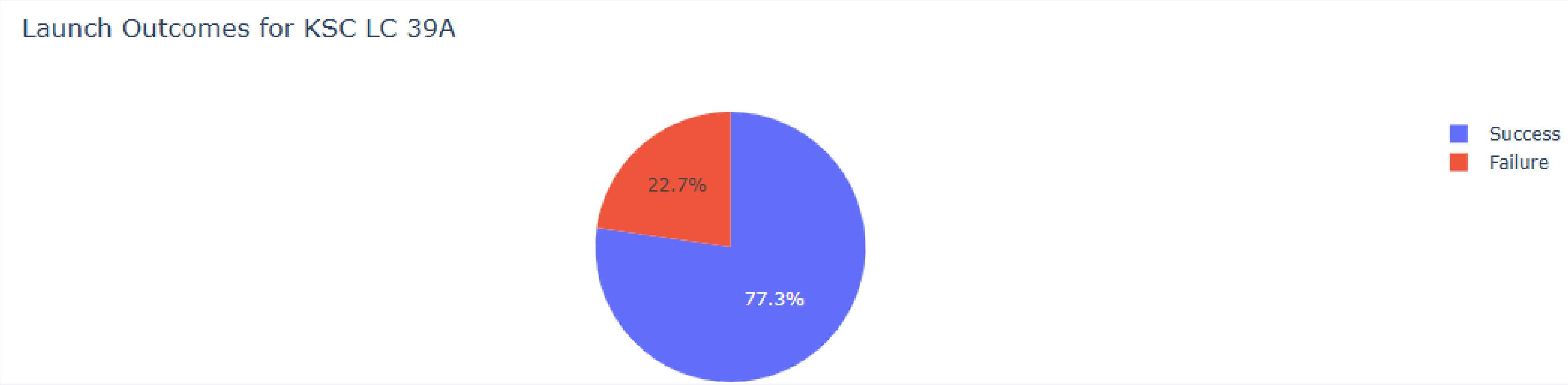
  The pie chart compares successful launches across SpaceX launch sites. Larger slices indicate sites with more successful launches and greater contributions to overall mission success. The distribution highlights the most active launch facilities and reveals differences in operational activity among sites.

<Dashboard Screenshot 2>

- Replace <Dashboard screenshot 2> title with an appropriate title

Launch Success Ratio for the Best-Performing Launch Site

- Show the screenshot of the piechart for the launch site with highest launch success ratio



- Explain the important elements and findings on the screenshot

The pie chart displays the proportion of successful and failed launches for the launch site with the highest success ratio.

Each slice represents a launch outcome category.

The success slice occupies the largest portion of the chart, indicating strong launch performance.

A small failure slice (or none at all) demonstrates a high mission success rate at this site.

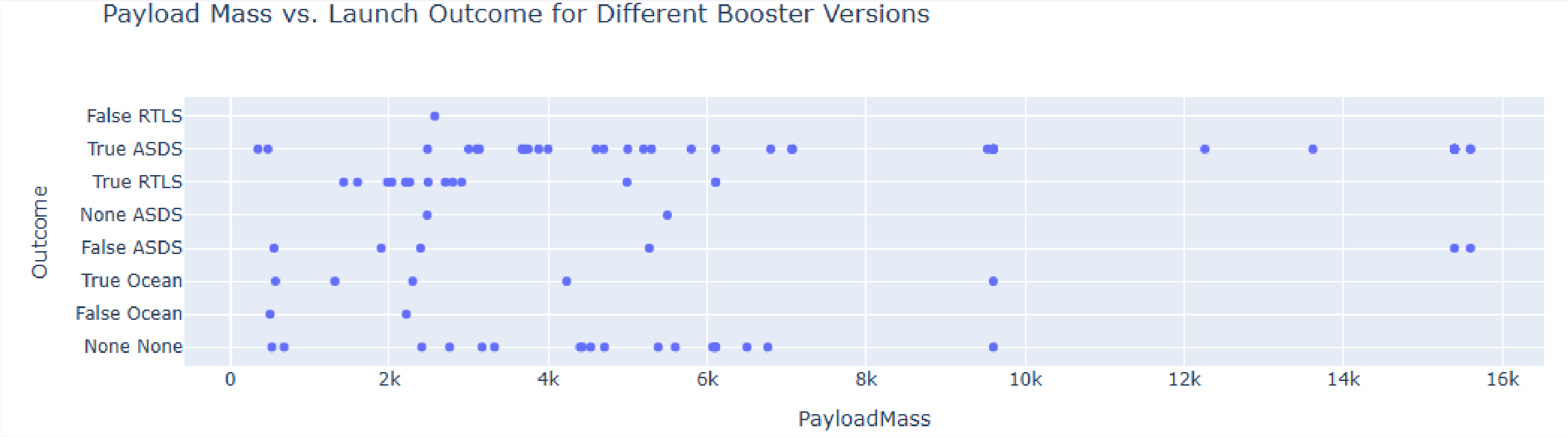
The visualization confirms that this launch site achieved the best operational performance among all SpaceX launch facilities.

The pie chart illustrates the distribution of successful and failed launches for the launch site with the highest success ratio. The dominance of the success segment highlights the site's excellent performance and reliability compared to other launch locations.

<Dashboard Screenshot 3>

- Replace <Dashboard screenshot 3> title with an appropriate title

- Show screenshots of Payload vs. Launch Outcome scatter plot for all sites, with different payload selected in the range slider



- Explain the important elements and findings on the screenshot, such as which payload range or booster version have the largest success rate, etc.

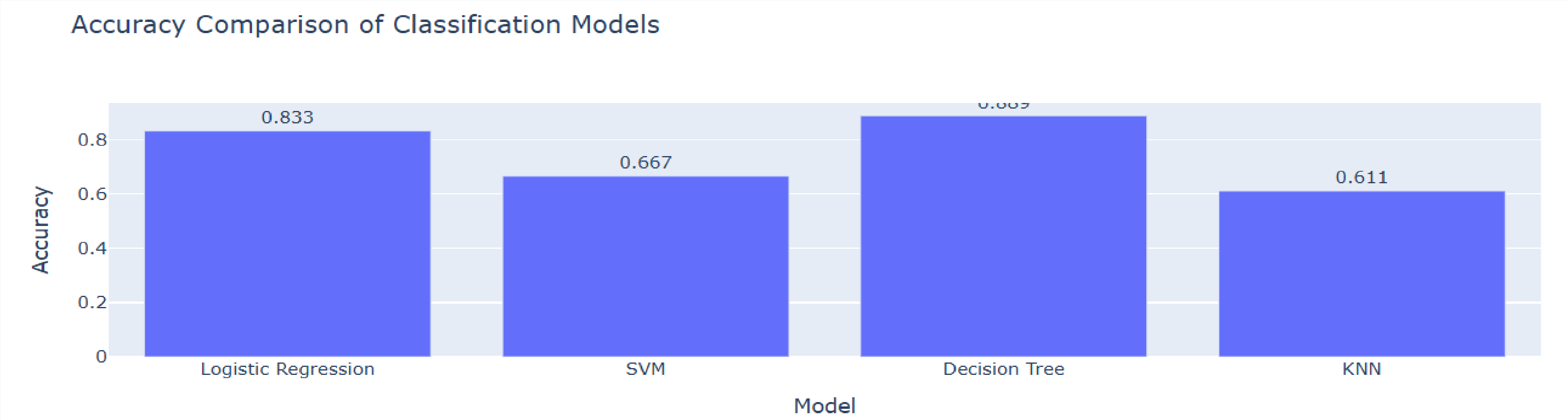
The scatter plot shows the relationship between payload mass and launch outcome. Successful launches are observed across a wide range of payload masses, indicating that payload mass alone is not a determining factor for mission success. When colored by booster version, the chart can also reveal which booster categories achieve higher success rates at different payload ranges.

Section 5

Predictive Analysis (Classification)

Classification Accuracy

- Visualize the built model accuracy for all built classification models, in a bar chart

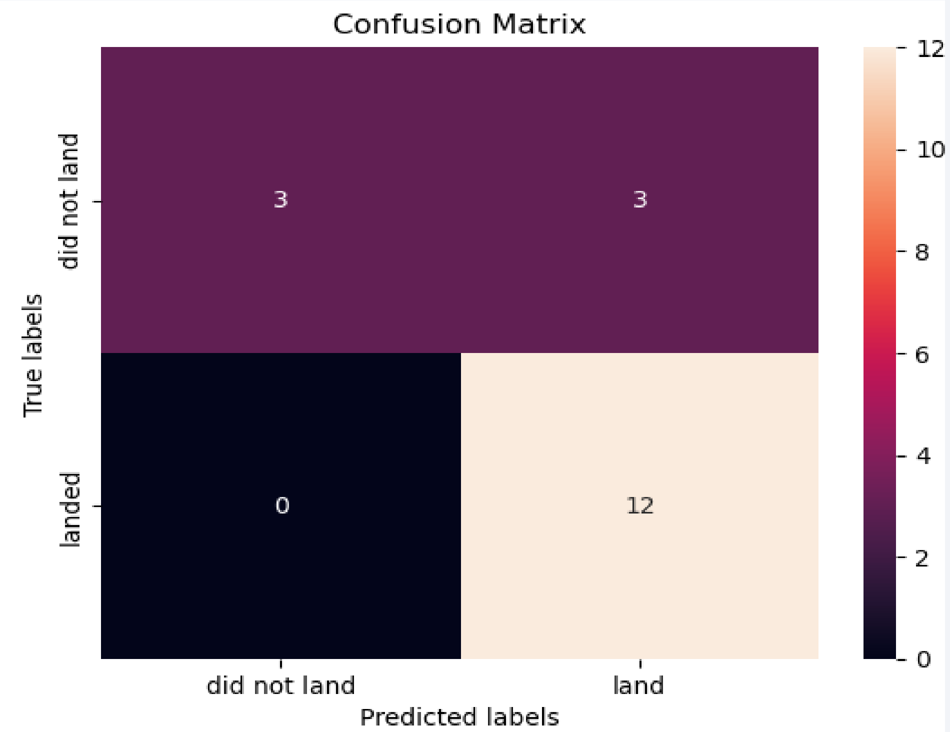


- Find which model has the highest classification accuracy

The Decision Tree model achieved the highest accuracy (88.8%), making it the best-performing classifier for predicting SpaceX launch outcomes. Logistic Regression also produced strong results with an accuracy of 83.3%, while SVM showed comparatively lower performance. The KNN accuracy value should be verified, as it does not correspond to a valid accuracy score.

Confusion Matrix

- Show the confusion matrix of the best performing model with an explanation



Conclusions

This project explored SpaceX launch data through data collection, exploratory analysis, interactive visualizations, geospatial analysis, and machine learning modeling. The analysis identified key factors influencing launch success, including launch site, payload mass, booster version, and geographical characteristics of launch facilities.

Interactive charts and maps provided valuable insights into launch success patterns and the proximity of launch sites to infrastructure such as coastlines, highways, and railways. Several classification models were then developed to predict launch outcomes.

Among the evaluated models, the Decision Tree classifier achieved the highest accuracy (88.8%), outperforming Logistic Regression (83.3%) and Support Vector Machine (66.7%). These results demonstrate that machine learning techniques can effectively predict launch success based on historical mission data.

Overall, this project highlights the value of data analytics and predictive modeling in understanding space launch operations and supporting data-driven decision-making in the aerospace industry.

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!